DAN ZHANG

Software Engineer • Computer Vision and Machine Learning
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PROFILE

Software engineer with around six years of industry experience in backend development, test automation and safety-critical embedded software, now completing an M.Sc. in Image Analysis and Machine Learning at Uppsala University (expected June 2027). At GE HealthCare, led development of a test automation platform that cut a recurring regression cycle from about one week to two days, and built backend features for an internal ML training platform for medical imaging. Recent projects span computer vision, data engineering and LLM-integrated applications, evaluated on real data. Available to start immediately for internships, part-time or full-time roles in Sweden.

PROFESSIONAL EXPERIENCE

GE HealthCare May 2021 – May 2025
Developer and Automated Test Engineer

- Led development of an internal UI-automation and test-execution platform with script management, real-time log monitoring, and automatic report generation, cutting a recurring regression test cycle from about one week to two days.
- Built backend features for an internal platform where ML teams ran medical imaging model training: authentication, user and group management, GPU/CPU resource allocation, training job queuing, and pipeline-triggered training runs, supporting image-quality and vessel-recognition models.
- Analyzed test logs and failures for root-cause investigation, wrote and reviewed quality documentation, and coordinated cross-team design reviews in a regulated healthcare setting.

Traffic Control Technology, China Aug 2018 – Mar 2020
Embedded Software Engineer

- Developed and unit-tested embedded C for safety-critical Automatic Train Operation (ATO) functions, including real-time speed control, acceleration/deceleration, station stopping, and sensor data processing.
- Designed and reviewed software interfaces to neighboring train-control systems (ATS, ZC) to support safe and reliable system integration.

China Everbright Bank Mar 2018 – Aug 2018
Software Engineer

- Contributed to the development and functional testing of a loan-verification system, implementing and validating business rules.

SELECTED PROJECTS

Drone detection and stereo 3D localization Uppsala University, 2026
Industry-partner team project, team of five • code on GitHub

- Fine-tuned YOLO26s on Anti-UAV300 using video-level splits. On 82 held-out stereo images from a separate drone flight, combined full-frame and tiled inference detected 73 of 78 annotated drones (5 missed, 1 false positive).
- Compared full-frame and overlapping-tile inference; extra fine-tuning on challenging backgrounds from the team's own flight increased false positives on validation, so kept the Anti-UAV300 model.
- Estimated camera poses, focal lengths and time offsets from DJI telemetry (PnP-RANSAC, robust least squares) and triangulated 3D positions: median difference 0.91 m on validation (33/41 pairs) and 1.18 m on a later same-flight interval (21/42).

SELECTED PROJECTS (CONTINUED)

LLM recipe assistant: data pipeline and recommendation service Uppsala University, 2026
Course project, team of three • code on GitHub

- Built a Python preprocessing pipeline for 13,480 recipes and 148,854 ingredient lines around a pretrained CRF ingredient parser, with cleaning, fallback handling and quality checks.
- Implemented a FastAPI service with REST endpoints for recipe ranking and missing-ingredient lookup, connected it to the LLM tool-calling workflow, and containerized it with Docker Compose.

Large-scale distributed text processing (Apache Spark / HDFS) Uppsala University, 2026
Data Engineering I, team project

- Designed and implemented the data ingestion and cleaning pipeline for a 6.4 GB Reddit dataset of 3,848,194 records, preparing JSONL and Parquet data for Spark and HDFS on a four-node OpenStack cluster.
- In the team's strong-scaling experiments, runtime fell from about 600 to 350 seconds when scaling from one to three workers; the analysis pointed to data shuffling and aggregation as likely bottlenecks.

EDUCATION

Uppsala University, Sweden 2025 – Jun 2027 (expected)
M.Sc. in Image Analysis and Machine Learning
Selected coursework: Data Engineering I (grade 5); Human-Robot Interaction (grade 5); Advanced Deep Learning for Image Processing (grade 4).

Beijing Technology and Business University, China Sep 2013 – Mar 2018
B.Sc. in Computer Science and Technology

TECHNICAL SKILLS

- **Programming:** Python, C, SQL, Java, JavaScript, Shell
- **Computer vision & machine learning:** PyTorch, Ultralytics YOLO, OpenCV, scikit-learn, NumPy, pandas; object detection, camera calibration, stereo triangulation, model evaluation
- **Backend & infrastructure:** FastAPI, REST APIs, Docker, Docker Compose, Git, Jenkins, Linux
- **Data engineering:** Apache Spark (PySpark), HDFS, JSONL, Parquet, OpenStack
- **Testing & quality:** test automation, regression testing, unit and functional testing, test-log and root-cause analysis
- **LLM applications:** LLM API integration, tool calling, database-grounded recommendation workflows

LANGUAGES

Chinese (native) • English (professional) • Swedish (basic, ongoing)